# RIJEKA

Open-Source Full Revaluation Risk System

## Quantitative Methodology & System Documentation

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|---|---|
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| **Primary Audience** | Model validators, quants, risk managers |

*This document is the single source of truth for all quantitative methodology, model validation support, and data lineage in Rijeka.*

***A feature is not production-ready until its methodology section is written.***

## Contents


**1 Purpose and Scope** **4**
1.1 Document Purpose . . . 4
1.2 ISDA SIMM as Universal Vocabulary . . . 4

**2 System Architecture Overview** **4**
2.1 Dual-Mode Computation Engine . . . 4
2.1.1 Mode 1: Full Revaluation . . . 4
2.1.2 Mode 2: Sensitivities (Taylor Approximation) . . . 5
2.2 Multi-Curve Framework . . . 5

**3 Market Data Architecture** **5**
3.1 Three-Store Design . . . 6
3.2 Data Quality Tier System . . . 6
3.3 VaR Readiness Check . . . 6

**4 Sprint 1: Market Data Workspace** **6**
4.1 Rates Curve Universe . . . 7
4.2 OIS Curve Framework . . . 7
4.2.1 Discount Factor and Zero Rate . . . 7
4.2.2 Bootstrap as Root-Finding Problem . . . 7
4.2.3 OIS Curve Conventions . . . 8
4.3 Interpolation Methods — Mathematical Specification . . . 8
4.3.1 1. Log-Linear Discount (Default) . . . 8
4.3.2 2. Linear Zero . . . 8
4.3.3 3. Cubic Spline Zero . . . 9
4.3.4 4. Log-Cubic Discount . . . 9
4.3.5 5. Monotone Cubic Zero (Hyman Filter) . . . 9
4.3.6 6. Flat Forward . . . 9
4.3.7 7. Convex Monotone . . . 9
4.3.8 8. Kruger Cubic Zero . . . 10
4.3.9 Interpolation Method Comparison . . . 10
4.4 Single-Currency Basis Curves . . . 10
4.4.1 Simultaneous Dual-Curve Bootstrap . . . 10
4.4.2 Tenor Basis Spreads . . . 11
4.5 Cross-Currency Basis Curves . . . 11
4.5.1 Covered Interest Parity . . . 11
4.5.2 XCCY Basis Swap Pricing . . . 11
4.5.3 XVA Implication — ColVA for Cross-Currency CSA . . . 11
4.6 Funding Curves . . . 12
4.6.1 Definition . . . 12
4.6.2 XVA Formulas — FVA, ColVA, MVA . . . 12
4.6.3 Sample Pre-Loaded Spreads . . . 13

**5 Planned Sections — Framework** **13**
5.1 Sprint 3: OIS Bootstrap API & History Store . . . 13
5.2 Sprint 4: Volatility Surfaces . . . 13
5.3 Sprints 7–8: Instrument Pricers . . . 14
5.4 Sprint 10: VaR Engine . . . 14

5.5 Sprint 15: XVA Engine . . . . . 14
5.6 Sprint 16: ISDA SIMM v2.6 . . . . . 15

**6 Model Validation Guide** **15**
6.1 OIS Bootstrap Validation Test Cases . . . . . 15
6.2 Known Model Limitations . . . . . 16

**7 Regulatory Alignment** **16**

**A Notation Reference** **17**

**B Document Version History** **17**

# 1 Purpose and Scope

## 1.1 Document Purpose

This document simultaneously serves five audiences:

| Purpose | Audience | Standard |
|---|---|---|
| Quantitative methodology | Quants, model validators | Industry best practice |
| Model validation support | MRM teams, counterparties | SR 11-7, SS1/23 |
| System configuration ref | Risk managers, technology | Internal governance |
| Data lineage documentation | Auditors, regulators | BCBS 239 |
| Regulatory alignment | Fed, ECB, FCA, CFTC | BCBS/IOSCO, EMIR |

## 1.2 ISDA SIMM as Universal Vocabulary

Rijeka uses ISDA SIMM v2.6 naming conventions throughout the entire system. Every risk class, sensitivity type, tenor, and sub-curve follows SIMM definitions, ensuring a model validator familiar with ISDA SIMM can immediately orient themselves without learning a parallel vocabulary.

**SIMM Sensitivity Notation**

$$s_{ik} \quad \text{Raw sensitivity of instrument } i \text{ to risk factor } k \tag{1}$$

$$WS_{ik} = RW_k \cdot s_{ik} \quad \text{(weighted sensitivity)} \tag{2}$$

$$K_b = \sqrt{\sum_i WS_{ib}^2 + \sum_{i \neq j} \rho_{ij}\, WS_{ib}\, WS_{jb}} \quad \text{(bucket margin)} \tag{3}$$

$$IM_{rc} = \sqrt{\sum_b K_b^2 + \sum_{b \neq c} \gamma_{bc}\, S_b\, S_c} \quad \text{(risk class margin)} \tag{4}$$

$$IM_{total} = \sum_{rc} IM_{rc} \quad \text{(total SIMM IM)} \tag{5}$$

where $\rho_{ij}$ is the within-bucket correlation and $\gamma_{bc}$ is the cross-bucket correlation from the SIMM v2.6 parameter set.

# 2 System Architecture Overview

## 2.1 Dual-Mode Computation Engine

Every analytics module supports two computation modes, selected at runtime.

### 2.1.1 Mode 1: Full Revaluation

For each trade $i$ and scenario $\omega$:

$$\Delta P_i(\omega) = \mathrm{NPV}_i(\mathcal{M}(\omega)) - \mathrm{NPV}_i(\mathcal{M}_0) \tag{6}$$

where $\mathcal{M}(\omega)$ is the full market data state under scenario $\omega$ and $\mathcal{M}_0$ is the base market data. Exact — captures all nonlinearity. Used for EOD, regulatory submissions, LCH IM, and backtesting.

### 2.1.2 Mode 2: Sensitivities (Taylor Approximation)

$$\Delta P \approx \underbrace{\sum_i s_i \, \Delta x_i}_{\text{delta}} + \underbrace{\frac{1}{2} \sum_{i,j} \Gamma_{ij} \, \Delta x_i \, \Delta x_j}_{\text{gamma}} + \underbrace{\mathcal{V} \, \Delta\sigma}_{\text{vega}} + \underbrace{\Theta \, \Delta t}_{\text{theta}} \tag{7}$$

Scenarios $\Delta \mathbf{x}$ are drawn from the SIMM covariance matrix:

$$\Sigma_{ij} = \sigma_i \, \sigma_j \, \rho_{ij} \tag{8}$$

where $\sigma_i = RW_i$ (SIMM risk weight, acting as volatility proxy) and $\rho_{ij}$ is the SIMM within-bucket correlation. Used for intraday monitoring, what-if analysis, and limit checking.

## 2.2 Multi-Curve Framework

Post-2008, the discount curve and projection curve are strictly separate objects.

**Definition**

**Discount curve:** the OIS curve for the CSA collateral currency.

$$D(t) = \exp(-r_{\text{OIS}}(t) \cdot t) \tag{9}$$

**Projection curve:** the index-specific forward curve.

$$F(t_1, t_2) = \frac{1}{\tau(t_1, t_2)} \left( \frac{D(t_1)}{D(t_2)} - 1 \right) \tag{10}$$

where $\tau(t_1, t_2)$ is the day count fraction under the relevant convention.

The NPV of a floating leg in currency $c$:

$$\text{NPV}_{\text{float}} = N \sum_{i=1}^{n} \tau_i \, F(t_{i-1}, \, t_i) \, D(t_i) \tag{11}$$

where $N$ is notional, $D(t_i)$ comes from the OIS curve (CSA currency), and $F(t_{i-1}, t_i)$ comes from the index projection curve (e.g. USD_TSOFR_3M for a Term SOFR 3M IRS).

# 3 Market Data Architecture

### 3.1 Three-Store Design

| Store | Purpose | Horizon | Mutability | Table |
|---|---|---|---|---|
| History | VaR, stress, LCH FHS, backtesting | Up to 10y (2520 bd) | **Immutable** — DB trigger blocks UPDATE/DELETE | `market_history` |
| Production | Committed EOD/intraday snapshots for all analytics | Rolling 90-day hot cache | Immutable once committed; amendment = new version | `production_snapshots` |
| Working | Live quotes, staging, what-if | Session only | Freely mutable; truncated each morning | `working_quotes` |

EOD commit is *atomic*: writes Production and History simultaneously. Both succeed or both roll back. The stores can never be out of sync.

### 3.2 Data Quality Tier System

Every data point carries an immutable quality tier stamped at ingestion. Risk engines enforce minimum tier requirements before any run.

| T | Name | Sources | Regulatory VaR |
|---|---|---|---|
| 1 | LIVE_VENDOR | Bloomberg BDP/BDH, Refinitiv | Permitted |
| 2 | OFFICIAL_CB | FRED, ECB, BOE, RBA, SNB, MAS | Permitted |
| 3 | VENDOR_IMPORT | Bloomberg CSV, Refinitiv CSV | Permitted |
| 4 | INTERNAL_IMPORT | Internal systems, broker axes | Permitted (with approval) |
| 5 | SYNTHETIC_SEED | Rijeka-generated (testing only) | **Blocked** |
| 6 | GAP_FILLED | Carry-forward / interpolation | **Blocked** |

### 3.3 VaR Readiness Check

Before any historical simulation, the system evaluates:

$$\text{coverage} = \frac{|\mathcal{D}_{\text{available}} \cap \mathcal{D}_{\text{required}}|}{|\mathcal{D}_{\text{required}}|} \tag{12}$$

where $\mathcal{D}_{\text{required}}$ is the set of required business dates (typically 1,250 days for VaR or 2,520 for LCH FHS) and $\mathcal{D}_{\text{available}}$ is the set of dates with at least Tier 1–4 data for all required risk factors. Regulatory runs are blocked if coverage $< 1.00$; indicative runs warn if coverage $< 0.99$.

## 4 Sprint 1: Market Data Workspace

**Sprint 1** **COMPLETE — OIS curves, surfaces, fixings, snap, snapshots**

### 4.1 Rates Curve Universe

| Class | Count | Description |
|---|---|---|
| OIS (INDEPENDENT) | 13 | DM overnight indexed swap mother curves |
| BASIS | 22 | Single-currency IBOR/term RFR vs OIS |
| XCCY_BASIS | 12 | Cross-currency basis — all DM pairs vs USD |
| FUNDING | 7 | Firm cost-of-funds: OIS + credit spread |
| **Total** | **54** | |

Bootstrap execution order follows topological sort of the dependency graph:

$$\underbrace{\text{Pass 1: OIS}}_{\text{independent}} \rightarrow \underbrace{\text{Pass 2: Basis}}_{\text{needs OIS}} \rightarrow \underbrace{\text{Pass 3: XCCY}}_{\text{needs OIS + FX}} \rightarrow \underbrace{\text{Pass 4: Funding}}_{\text{needs OIS, no solver}} \tag{13}$$

### 4.2 OIS Curve Framework

#### 4.2.1 Discount Factor and Zero Rate

**Fundamental Curve Relations**

Let $D(t)$ denote the discount factor for maturity $t$ (in years). The continuously compounded zero rate is:

$$r(t) = -\frac{\ln D(t)}{t} \qquad \Longleftrightarrow \qquad D(t) = e^{-r(t)\,t} \tag{14}$$

The instantaneous forward rate:

$$f(t) = -\frac{\partial}{\partial t} \ln D(t) = r(t) + t\,\frac{\partial r}{\partial t} \tag{15}$$

The simply compounded period forward rate for $[t_1, t_2]$:

$$F(t_1, t_2) = \frac{1}{\tau(t_1, t_2)} \left( \frac{D(t_1)}{D(t_2)} - 1 \right) \tag{16}$$

The par OIS swap rate for maturity $T$ with fixed payment times $\{t_i\}$:

$$S(T) = \frac{D(0) - D(T)}{\sum_{i=1}^{n} \tau_i\, D(t_i)} = \frac{1 - D(T)}{A(T)} \tag{17}$$

where $A(T) = \sum_{i=1}^{n} \tau_i D(t_i)$ is the annuity (DV01 numeraire).

#### 4.2.2 Bootstrap as Root-Finding Problem

Given market quotes $\{(T_j,\, Q_j)\}_{j=1}^{n}$ — where $Q_j$ is the observed par OIS swap rate for maturity $T_j$ — the bootstrap solves for discount factors $\{D(T_j)\}$ such that each instrument reprices to par:

$$\mathcal{F}_j(D(T_1), \ldots, D(T_j)) = Q_j^{\text{model}} - Q_j^{\text{market}} = 0 \qquad j = 1, \ldots, n \tag{18}$$

For an OIS swap this gives an *exact* closed-form solution for $D(T_j)$ given $D(T_1), \ldots, D(T_{j-1})$ (sequential bootstrap, no iteration needed for piecewise-constant forwards). QuantLib uses Brent's method for general interpolation methods where no closed form exists:

$$D(T_j) = \arg\min_D \left| S^{\text{model}}(T_j;\, D) - Q_j \right| \tag{19}$$

with convergence tolerance $\varepsilon = 10^{-12}$ on the rate residual.

### 4.2.3 OIS Curve Conventions

| Ccy | Index | QL Class | Day Count | Calendar | T+ | Lag | Tele. |
|---|---|---|---|---|---|---|---|
| USD | SOFR | `ql.Sofr()` | Act/360 | US(FedRes) | 1 | 2 | Yes |
| EUR | €STR | `ql.Estr()` | Act/360 | TARGET() | 2 | 2 | Yes |
| GBP | SONIA | `ql.Sonia()` | Act/365F | UK(Exchange) | 0 | 0 | No |
| JPY | TONAR | `ql.Tonar()` | Act/365F | Japan() | 2 | 2 | Yes |
| CHF | SARON | `ql.Saron()` | Act/360 | Switzerland() | 2 | 2 | Yes |
| AUD | AONIA | `ql.Aonia()` | Act/365F | Australia() | 0 | 0 | Yes |
| CAD | CORRA | `ql.Corra()` | Act/365F | Canada() | 1 | 2 | Yes |
| SEK | SWESTR | `OvernightIndex` | Act/360 | Sweden() | 1 | 1 | Yes |
| NOK | NOWA | `OvernightIndex` | Act/360 | Norway() | 1 | 1 | Yes |
| DKK | DESTR | `OvernightIndex` | Act/360 | Denmark() | 1 | 1 | Yes |
| NZD | NZIONA | `OvernightIndex` | Act/365F | NZ() | 0 | 0 | Yes |
| HKD | HONIA | `OvernightIndex` | Act/365F | HongKong() | 0 | 0 | Yes |
| SGD | SORA | `OvernightIndex` | Act/365F | Singapore() | 0 | 0 | Yes |

**GBP SONIA Convention Note**

GBP SONIA is unique among the DM OIS curves: settlement is T+0 and payment lag is 0 days. This reflects Bank of England convention. In QuantLib, set `telescopicValueDates=False` for SONIA — all other DM OIS curves use `True`.

## 4.3 Interpolation Methods — Mathematical Specification

Eight interpolation methods are supported. The choice determines the functional form of $D(t)$ (or $r(t)$ or $f(t)$) between pillar dates. Let $\{t_0 < t_1 < \cdots < t_n\}$ be the bootstrap pillars.

### 4.3.1 1. Log-Linear Discount (Default)

Interpolates $\ln D(t)$ linearly between pillars:

$$\ln D(t) = \ln D(t_i) + \frac{t - t_i}{t_{i+1} - t_i} \left[\ln D(t_{i+1}) - \ln D(t_i)\right], \qquad t \in [t_i, t_{i+1}] \tag{20}$$

$$\implies \quad f(t) = \frac{\ln D(t_i) - \ln D(t_{i+1})}{t_{i+1} - t_i} = \text{const on } [t_i, t_{i+1}] \tag{21}$$

Instantaneous forward rates are *piecewise constant* and *positive* by construction. Industry standard for OIS bootstrapping. QuantLib class: `PiecewiseLogLinearDiscount`.

### 4.3.2 2. Linear Zero

Interpolates the zero rate $r(t)$ linearly:

$$r(t) = r(t_i) + \frac{t - t_i}{t_{i+1} - t_i} \left[r(t_{i+1}) - r(t_i)\right] \tag{22}$$

Simple and fast. Instantaneous forwards are piecewise constant (same structure as log-linear discount but on rates, not log-DFs). QuantLib: `PiecewiseLinearZero`.

#### 4.3.3 3. Cubic Spline Zero

Interpolates $r(t)$ with a natural cubic spline. The spline satisfies:

$$r(t) = a_i + b_i(t - t_i) + c_i(t - t_i)^2 + d_i(t - t_i)^3, \qquad t \in [t_i, t_{i+1}] \tag{23}$$

with $C^2$ continuity: $r$, $r'$, and $r''$ all continuous at each pillar. Natural boundary conditions: $r''(t_0) = r''(t_n) = 0$. Produces smooth, continuous forward rates. Risk: Runge phenomenon at endpoints. QuantLib: `PiecewiseCubicZero`.

The instantaneous forward rate is:

$$f(t) = r(t) + t\,r'(t) = a_i + (2b_i + a_i/t)(t - t_i) + \cdots \tag{24}$$

where $r'(t)$ is the spline derivative — smooth everywhere.

#### 4.3.4 4. Log-Cubic Discount

Interpolates $\ln D(t)$ with a cubic spline (same $C^2$ conditions as above but applied to the log-discount):

$$\ln D(t) = a_i + b_i(t - t_i) + c_i(t - t_i)^2 + d_i(t - t_i)^3 \tag{25}$$

$$f(t) = -\frac{\partial}{\partial t} \ln D(t) = -(b_i + 2c_i(t - t_i) + 3d_i(t - t_i)^2) \tag{26}$$

Forwards are smooth but not guaranteed positive. Good balance of smoothness and stability. QuantLib: `PiecewiseLogCubicDiscount`.

#### 4.3.5 5. Monotone Cubic Zero (Hyman Filter)

Cubic spline on $r(t)$ with Hyman (1983) monotonicity filter applied to spline coefficients. For each interval, the filter constrains $c_i$ and $d_i$ such that the interpolant preserves local monotonicity:

$$\text{if } r(t_i) \le r(t_{i+1}) \text{ then } r(t) \text{ is non-decreasing on } [t_i, t_{i+1}] \tag{27}$$

The filter replaces $c_i$ with:

$$\tilde{c}_i = \min\left(3\,\frac{r(t_{i+1}) - r(t_i)}{h_i},\ c_i\right) \tag{28}$$

Eliminates oscillations without global dampening. QuantLib: `PiecewiseMonotoneCubicZero`.

#### 4.3.6 6. Flat Forward

Interpolates the instantaneous forward rate $f(t)$ as piecewise constant, solving directly for the forward level that reprices each instrument:

$$f(t) = f_i = \text{const}, \quad t \in (t_{i-1}, t_i] \tag{29}$$

$$D(t) = D(t_{i-1})\,\exp(-f_i\,(t - t_{i-1})) \tag{30}$$

Maximum locality — changing one pillar affects only the adjacent segment. QuantLib: `PiecewiseFlatForward`.

#### 4.3.7 7. Convex Monotone

Andersen & Piterbarg (2010) convex monotone interpolation on $f(t)$. Parameterises the forward curve as:

$$f(t) = \phi_i + \psi_i\, g\left(\frac{t - t_{i-1}}{h_i}\right), \quad g(x) = \frac{x^2 - x}{2}, \quad h_i = t_i - t_{i-1} \tag{31}$$

where $\phi_i$, $\psi_i$ are chosen to ensure $f(t) > 0$ everywhere and convexity is preserved between pillars. QuantLib: `PiecewiseConvexMonotone`.

#### 4.3.8 8. Kruger Cubic Zero

Kruger (2002) locally-monotone cubic interpolation on $r(t)$. Unlike the global Hyman filter, Kruger's algorithm is *local*: it computes each spline slope from adjacent data only, avoiding the global dampening effect:

$$r'(t_i) = \begin{cases} \frac{h_{i-1}\,\delta_i + h_i\,\delta_{i-1}}{h_{i-1}+h_i} & \text{if } \delta_{i-1}\,\delta_i > 0 \\ 0 & \text{otherwise} \end{cases} \tag{32}$$

where $\delta_i = (r(t_{i+1}) - r(t_i))/h_i$ is the divided difference. QuantLib: `PiecewiseKrugerZero`.

#### 4.3.9 Interpolation Method Comparison

| Method | Interpolates | Forward | Positive $f$? | Best For |
|---|---|---|---|---|
| Log-Linear Discount | $\ln D(t)$ | Piecewise const | **Guaranteed** | OIS — **Default** |
| Linear Zero | $r(t)$ | Piecewise const | Not guaranteed | Simple structures |
| Cubic Spline Zero | $r(t)$ | Smooth ($C^1$) | Not guaranteed | Smooth fwd profile |
| Log-Cubic Discount | $\ln D(t)$ | Smooth ($C^1$) | Not guaranteed | Gov't curves |
| Monotone Cubic | $r(t)$ | Smooth, monotone | Conditionally | Curves without oscillation |
| Flat Forward | $f(t)$ | Step function | **Guaranteed** | Max locality |
| Convex Monotone | $f(t)$ | Smooth, convex | **Guaranteed** | Central bank curves |
| Kruger Cubic | $r(t)$ | Smooth, local | Conditionally | Alt. to Hyman filter |

### 4.4 Single-Currency Basis Curves

**Sprint 3** **PLANNED — Bootstrap wired Sprint 3**

#### 4.4.1 Simultaneous Dual-Curve Bootstrap

The basis curve is not bootstrapped independently. It is solved simultaneously with the OIS discount curve via the following system. Let $D^{\text{OIS}}(t)$ denote the already-bootstrapped OIS discount factors. The basis projection curve $D^{\text{basis}}(t)$ is found by solving:

$$S^{\text{IRS}}(T) = \frac{\sum_{i=1}^{n^f} \tau_i^f\, F^{\text{basis}}(t_{i-1}^f, t_i^f)\, D^{\text{OIS}}(t_i^f)}{\sum_{j=1}^{n^f} \tau_j^f\, D^{\text{OIS}}(t_j^f)} = Q_T \tag{33}$$

where $F^{\text{basis}}$ is the forward rate implied by $D^{\text{basis}}$:

$$F^{\text{basis}}(t_1, t_2) = \frac{1}{\tau(t_1, t_2)} \left( \frac{D^{\text{basis}}(t_1)}{D^{\text{basis}}(t_2)} - 1 \right) \tag{34}$$

The OIS curve provides discounting; the basis curve provides projection. They are decoupled — $D^{\text{OIS}}$ enters as a fixed input to the basis bootstrap.

### 4.4.2 Tenor Basis Spreads

For a tenor basis swap (e.g. 3M vs 1M SOFR), the market quote is the spread $s$ such that:

$$\sum_i \tau_i^{3M} F^{3M}(t_{i-1}, t_i)\, D(t_i) = \sum_j \tau_j^{1M} \left[F^{1M}(t_{j-1}, t_j) + s\right] D(t_j) \tag{35}$$

Rearranging, the implied spread between the two projection curves is:

$$s = \frac{\sum_i \tau_i^{3M} F_i^{3M} D_i - \sum_j \tau_j^{1M} F_j^{1M} D_j}{\sum_j \tau_j^{1M} D_j} \tag{36}$$

Typical ranges: USD SOFR 1M/3M: $-2$ to $-5$ bp; EUR EURIBOR 1M/3M: $-3$ to $-8$ bp.

## 4.5 Cross-Currency Basis Curves

**Sprint 6** **PLANNED — Bootstrap wired Sprint 6**

### 4.5.1 Covered Interest Parity

In a frictionless market, the forward FX rate is determined by the interest rate differential (Covered Interest Parity, CIP):

$$F^{\mathrm{CIP}}(t) = S_0 \cdot \frac{D^{\mathrm{dom}}(t)}{D^{\mathrm{USD}}(t)} \tag{37}$$

where $S_0$ is the FX spot rate (DOM/USD) and $D^{\mathrm{dom}}$, $D^{\mathrm{USD}}$ are the domestic and USD OIS discount factors.

Post-2008, CIP fails in practice. The observed forward $F^{\mathrm{mkt}}(t)$ deviates from $F^{\mathrm{CIP}}(t)$. The cross-currency basis $b(t)$ captures this deviation:

$$F^{\mathrm{mkt}}(t) = S_0 \cdot \frac{D^{\mathrm{dom}}(t)\, e^{b(t)\, t}}{D^{\mathrm{USD}}(t)} \tag{38}$$

The XCCY bootstrap solves for $b(t)$ at each pillar such that a mark-to-market XCCY basis swap (exchanging domestic OIS compounded vs USD OIS compounded) prices to par given observed market quotes.

### 4.5.2 XCCY Basis Swap Pricing

A receive-DOM / pay-USD XCCY basis swap with basis spread $b$ on the DOM leg:

$$\mathrm{NPV} = \underbrace{N_{\mathrm{DOM}} \left[\sum_i \tau_i \left(F_i^{\mathrm{OIS,dom}} + b\right) D_i^{\mathrm{dom}} + D^{\mathrm{dom}}(T) - 1\right]}_{\text{domestic leg (in DOM ccy)}} - \underbrace{N_{\mathrm{USD}} \frac{1}{S_0} \left[\sum_i \tau_i F_i^{\mathrm{OIS,USD}} D_i^{\mathrm{USD}} + D^{\mathrm{USD}}(T) - 1\right]}_{\text{USD leg (converted to DOM)}} \tag{39}$$

The notionals satisfy $N_{\mathrm{DOM}} = N_{\mathrm{USD}}/S_0$.

### 4.5.3 XVA Implication — ColVA for Cross-Currency CSA

When a USD trade is collateralised under a EUR CSA, the cheapest-to-deliver (CTD) discount rate involves the XCCY basis:

$$r_{\mathrm{CTD}}(t) = r^{€\mathrm{STR}}(t) + b_{\mathrm{EUR/USD}}(t) \tag{40}$$

The resulting ColVA adjustment:

$$\mathrm{ColVA} = -\int_0^T D(t)\,\mathbb{E}^{\mathbb{Q}}[\mathrm{EE}(t)] \cdot \big(r_{\mathrm{CTD}}(t) - r_{\mathrm{OIS,USD}}(t)\big)\ \mathrm{d}t \tag{41}$$

*Without the XCCY curve, this calculation is incorrect for cross-currency CSA trades.*

## 4.6 Funding Curves

**Sprint 1** **COMPLETE — Manual spread UI built. XVA linkage defined.**

### 4.6.1 Definition

**Funding Curve**

The firm's funding curve at tenor $t$ is:

$$r_{\mathrm{fund}}(t) = r_{\mathrm{OIS}}(t) + \frac{s(t)}{10\,000} \tag{42}$$

where $s(t)$ is the firm's credit spread above OIS in basis points. The spread $s(t)$ is a term structure input manually by the treasury desk; it is not bootstrapped from market instruments. Interpolation between input tenors: log-linear, consistent with the OIS base curve.

This is not bootstrapped because no directly observable market instruments constrain the firm's funding spread term structure. The OIS bootstrap (Pass 1) provides the risk-free base; the spread is purely additive.

### 4.6.2 XVA Formulas — FVA, ColVA, MVA

$$\mathrm{FVA} = -\int_0^T D(t)\,\mathbb{E}^{\mathbb{Q}}[\mathrm{EE}(t) - \mathrm{ENE}(t)] \cdot \frac{s(t)}{10\,000}\ \mathrm{d}t \tag{43}$$

$$\mathrm{ColVA} = -\int_0^T D(t)\,\mathbb{E}^{\mathbb{Q}}[\mathrm{EE}(t)] \cdot s_{\mathrm{CTD}}(t)\ \mathrm{d}t \tag{44}$$

$$\mathrm{MVA} = -\int_0^T D(t)\,\mathbb{E}^{\mathbb{Q}}[\mathrm{SIMM\text{-}IM}(t)] \cdot \frac{s(t)}{10\,000}\ \mathrm{d}t \tag{45}$$

where:

- $D(t)$ — OIS discount factor (from base curve, Pass 1)
- $\mathrm{EE}(t)$ — Expected Exposure under risk-neutral measure $\mathbb{Q}$
- $\mathrm{ENE}(t)$ — Expected Negative Exposure
- $s(t)$ — firm credit spread in bp (from funding curve)
- $s_{\mathrm{CTD}}(t)$ — cheapest-to-deliver collateral spread
- $\mathrm{SIMM\text{-}IM}(t)$ — projected forward SIMM IM path (Sprint 16 input)

In discrete form (for numerical implementation, Sprint 15):

$$\mathrm{FVA} \approx -\sum_{k=1}^{M} D(t_k)\left[\mathrm{EE}(t_k) - \mathrm{ENE}(t_k)\right] \cdot \frac{s(t_k)}{10\,000} \cdot \Delta t_k \tag{46}$$

$$\mathrm{MVA} \approx -\sum_{k=1}^{M} D(t_k)\,\mathrm{SIMM\text{-}IM}(t_k) \cdot \frac{s(t_k)}{10\,000} \cdot \Delta t_k \tag{47}$$

### 4.6.3 Sample Pre-Loaded Spreads

Representative of a Tier 1, A-rated bank. Users replace with own institution's spreads.

| Tenor | USD | EUR | GBP | JPY | CHF | AUD | CAD |
|---|---|---|---|---|---|---|---|
| ON | 12 | 14 | 10 | 5 | 10 | 15 | 12 |
| 1M | 20 | 22 | 18 | 8 | 15 | 22 | 18 |
| 3M | 28 | 30 | 25 | 12 | 22 | 30 | 25 |
| 6M | 35 | 38 | 32 | 16 | 28 | 38 | 32 |
| 1Y | 42 | 45 | 40 | 20 | 35 | 45 | 40 |
| 2Y | 52 | 55 | 50 | 25 | 42 | 55 | 50 |
| 3Y | 58 | 62 | 55 | 28 | 48 | 60 | 55 |
| 5Y | 65 | 70 | 62 | 32 | 55 | 68 | 62 |
| 7Y | 70 | 75 | 68 | 36 | 60 | — | — |
| 10Y | 75 | 80 | 72 | 38 | 62 | 75 | 70 |
| 20Y | 85 | 90 | 82 | 42 | — | — | — |
| 30Y | 88 | 92 | 85 | 44 | — | — | — |

All values in basis points above OIS. Blank = pillar not defined.

## 5 Planned Sections — Framework

### 5.1 Sprint 3: OIS Bootstrap API & History Store

**Sprint 3** **PLANNED**

Will document: the live QuantLib bootstrap API endpoint, simultaneous basis curve bootstrap implementation, three-store schema deployment, Bloomberg DAPI adapter, FRED/ECB/BOE adapters. Mathematical content: convergence proof sketch for Brent's method applied to OIS bootstrap; sensitivity of bootstrap result to pillar instrument selection; condition number analysis.

### 5.2 Sprint 4: Volatility Surfaces

**Sprint 4** **PLANNED**

Will document SABR calibration. The SABR model (Hagan et al. 2002) for the implied vol smile:

$$\sigma_{\mathrm{BS}}(F,K,T) \approx \frac{\alpha}{(FK)^{\frac{1-\beta}{2}}} \cdot \frac{z}{\chi(z)} \cdot \left[1 + \left(\frac{(1-\beta)^2}{24}\frac{\alpha^2}{(FK)^{1-\beta}} + \frac{\rho\beta\nu\alpha}{4(FK)^{\frac{1-\beta}{2}}} + \frac{2-3\rho^2}{24}\nu^2\right)T\right] \tag{48}$$

where $z = \frac{\nu}{\alpha}(FK)^{\frac{1-\beta}{2}}\ln(F/K)$ and $\chi(z) = \ln\left(\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}\right)$.

Parameters $(\alpha, \beta, \rho, \nu)$ calibrated per expiry slice by minimising the weighted sum of squared vega-normalised errors across the smile.

### 5.3 Sprints 7–8: Instrument Pricers

**Sprint 7** **PLANNED**

Will document NPV formulas for: IRS, OIS swap, basis swap, XCCY swap, swaption (Black and Bachelier models), cap/floor, CDS (ISDA model), CDX index.

For reference, the key pricing equations are:

$$\mathrm{NPV}_{\mathrm{IRS}} = N\left[\sum_i \tau_i^f F_i^{\mathrm{basis}} D_i - K \sum_j \tau_j^f D_j\right] \tag{49}$$

$$\mathrm{NPV}^{\mathrm{Black}}_{\mathrm{swaption}} = A(T_0)\left[S(T_0,T_n)\,\Phi(d_+) - K\,\Phi(d_-)\right] \tag{50}$$

$$d_\pm = \frac{\ln(S/K) \pm \frac{1}{2}\sigma^2 T_0}{\sigma\sqrt{T_0}} \tag{51}$$

### 5.4 Sprint 10: VaR Engine

**Sprint 10** **PLANNED**

**Historical Simulation VaR** (1-day, 99%, 1250 scenarios):

$$\mathrm{P\&L}_t = \sum_i \mathrm{NPV}_i(\mathcal{M}_t) - \sum_i \mathrm{NPV}_i(\mathcal{M}_{t-1}) \tag{52}$$

$$\mathrm{VaR}_{0.99} = -\mathrm{Percentile}\big(\{\mathrm{P\&L}_t\}_{t=1}^{1250},\ 1\%\big) \tag{53}$$

**Delta-Gamma VaR** (using SIMM covariance):

$$\mathrm{VaR}_{\Delta\Gamma} \approx z_{0.99}\sqrt{\mathbf{WS}^{\mathrm{T}}\,\Sigma\,\mathbf{WS}} + \frac{1}{2}\,\mathrm{tr}(\Gamma\,\Sigma) \tag{54}$$

**Expected Shortfall** (for FRTB):

$$\mathrm{ES}_{0.975} = -\frac{1}{n\,(1-0.975)} \sum_{t:\,\mathrm{P\&L}_t < \mathrm{VaR}_{0.975}} \mathrm{P\&L}_t \tag{55}$$

### 5.5 Sprint 15: XVA Engine

**Sprint 15** **PLANNED**

Full bilateral XVA:

$$\mathrm{CVA} = -\mathrm{LGD}_c \int_0^T D(t)\,\mathbb{E}^{\mathbb{Q}}[\mathrm{EE}(t)]\,\mathrm{dPD}_c^{\mathbb{Q}}(t) \tag{56}$$

$$\mathrm{DVA} = +\mathrm{LGD}_b \int_0^T D(t)\,\mathbb{E}^{\mathbb{Q}}[\mathrm{ENE}(t)]\,\mathrm{dPD}_b^{\mathbb{Q}}(t) \tag{57}$$

$$\mathrm{KVA} = -\int_0^T D(t)\,\mathbb{E}^{\mathbb{Q}}[\mathrm{EAD}(t)] \cdot RW \cdot \mathrm{CoC}\ \mathrm{d}t \tag{58}$$

FVA and MVA are given by equations (43) and (45).

### 5.6 Sprint 16: ISDA SIMM v2.6

**Sprint 16** **PLANNED**

Implemented entirely in own Python. The complete aggregation:

$$WS_{ik} = RW_k \cdot s_{ik} \tag{59}$$

$$K_b = \sqrt{\sum_i WS_{ib}^2 + \sum_{i \neq j} \rho_{ij}\, WS_{ib}\, WS_{jb}} \tag{60}$$

$$S_b = \max\left(-K_b,\ \min\left(K_b,\ \sum_i WS_{ib}\right)\right) \tag{61}$$

$$IM_{rc} = \sqrt{\sum_b K_b^2 + \sum_{b \neq c} \gamma_{bc}\, S_b\, S_c} \tag{62}$$

$$IM_{\text{total}} = \sum_{rc} IM_{rc} \tag{63}$$

## 6 Model Validation Guide

### 6.1 OIS Bootstrap Validation Test Cases

| Test ID | Input | Expected Output | Tolerance |
|---|---|---|---|
| OIS-V001 | USD SOFR: ON=5.31%, 1Y=4.48%, 5Y=3.78%, 10Y=3.74% | $D(1) \approx 0.9571$, $D(5) \approx 0.8289$, $D(10) \approx 0.6898$ | $10^{-4}$ on $D$ |
| OIS-V002 | EUR €STR: ON=3.40%, 1Y=2.58%, 5Y=2.30% | $r(5) \approx 2.283\%$, $r(10) \approx 2.448\%$ | 0.1 bp on $r$ |
| OIS-V003 | Flat curve: all tenors $= r$ | $D(t) = e^{-rt}$ exactly for all $t$ | Machine $\varepsilon$ |
| OIS-V004 | Log-linear vs Cubic Spline at 3.5Y | Forward rate difference $< 20$ bp for normal curve shapes | Qualitative |

### 6.2 Known Model Limitations

| Model | Limitation | Impact | Mitigation |
|---|---|---|---|
| All interpolation | Assumes continuous trading at pillar tenors; illiquid pillars ill-conditioned | Curve poorly defined at illiquid points | Illiquid pillars (3W, 4Y, 15Y) disabled by default |
| Log-Linear Discount | Piecewise-constant forwards; unrealistic rate expectations | Stepped forward curve; affects swaption pricing at boundaries | Switch to Cubic or Monotone Cubic for smooth forwards |
| Bootstrap accuracy $10^{-12}$ | Will not converge for inconsistent or highly non-monotone instruments | Bootstrap fails — exception raised | Pre-validation range checks (Sprint 3) |
| Flat extrapolation | Beyond last pillar, uses last-pillar rate | Inaccurate for ultra-long trades ($> 30$Y) | User can add longer pillar |

## 7 Regulatory Alignment

| Requirement | Rijeka Module | Sprint | Standard |
|---|---|---|---|
| IM — SIMM | SIMM engine, all 6 risk classes | 16 | ISDA SIMM v2.6 |
| IM — Schedule | Schedule IM engine | 17 | BCBS/IOSCO WGMR |
| Variation Margin | Full revaluation MTM | 3 | BCBS/IOSCO WGMR |
| CCR Capital | SA-CCR EAD engine | 14 | BCBS d424 |
| Market Risk | FRTB ES (2520-day window) | 10 | BCBS d457 |
| Data Lineage | Immutable history + audit trail | 3 | BCBS 239 |
| LCH SwapClear | FHS 1250-day replication | 17 | LCH methodology |

## A Notation Reference

| Symbol | Definition |
|---|---|
| $D(t)$ | Discount factor for maturity $t$: present value of 1 unit at time $t$ |
| $r(t)$ | Continuously compounded zero rate |
| $f(t)$ | Instantaneous forward rate |
| $F(t_1, t_2)$ | Simply compounded period forward rate for $[t_1, t_2]$ |
| $S(T)$ | Par swap rate for maturity $T$ |
| $A(T)$ | Annuity (DV01 numeraire): $\sum_i \tau_i D(t_i)$ |
| $\tau(t_1, t_2)$ | Day count fraction under the relevant convention |
| $N$ | Notional |
| $K$ | Fixed rate (IRS) or strike (option) |
| $\Phi(\cdot)$ | Standard normal CDF |
| $\mathbb{E}^{\mathbb{Q}}[\cdot]$ | Expectation under risk-neutral measure $\mathbb{Q}$ |
| $\mathrm{EE}(t)$ | Expected Exposure at time $t$ |
| $\mathrm{ENE}(t)$ | Expected Negative Exposure |
| $\mathrm{LGD}$ | Loss Given Default |
| $\mathrm{PD}(t)$ | Default probability (survival: $1 - \mathrm{PD}$) |
| $s(t)$ | Firm credit spread at tenor $t$ (basis points) |
| $s_{ik}$ | SIMM raw sensitivity of instrument $i$ to factor $k$ |
| $RW_k$ | SIMM risk weight for factor $k$ |
| $\rho_{ij}$ | SIMM within-bucket correlation |
| $\gamma_{bc}$ | SIMM cross-bucket correlation |
| $\Sigma$ | SIMM covariance matrix: $\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$ |

## B Document Version History

| Version | Date | Author | Changes |
|---|---|---|---|
| v1.0 | 2025-01-15 | Rijeka Core Team | Initial release. Sprint 1 complete. Framework for Sprints 2–20. |
| v1.1 | 2025-01-15 | Rijeka Core Team | Rebuilt in LaTeX. Full mathematical notation throughout. Sections 4.3 (Basis), 4.4 (XCCY), 4.5 (Funding) added. All formulas typeset. |

**RIJEKA** — Open-Source Full Revaluation Risk System

*One system. One vocabulary. Every risk metric a derivatives book requires.*